Avishkar Mahesh Pawar

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Portfolio GitHub LinkedIn

Research Interests

Distributed storage and data systems for scalable AI workloads; database internals; transaction and concurrency control; consensus and replication; streaming systems; vector search; retrieval-augmented generation infrastructure.

Education

MIT Academy of Engineering, Pune 2022–2026
B.Tech. in Computer Engineering CGPA: 9.24/10

Examinations

GATE CS 2026 Qualified AIR 3124, Score 601, Marks 51.45/100
GATE CS 2025 Qualified AIR 9860, Score 453

Selected Projects

MiniDB – Distributed Database C++17, CMake, Google Test, POSIX Sockets
github.com/avishkar-004/minidb

- Built a distributed key-value database with B+ Tree indexing, MVCC transactions, Raft consensus, consistent hash sharding, SQL-like query processing with JOINs, and HTTP monitoring.
- Implemented core database internals including indexing, transaction handling, replication, query execution, and storage coordination.
- Reported performance: 10,000+ ops/sec, sub-1ms latency in project benchmarks.

StreamFlow – Event Streaming Platform Java 17, Spring Boot, Netty, Docker, Prometheus
github.com/avishkar-004/streamflow

- Built a Kafka-like distributed message queue with consumer groups, leader-follower replication, ISR tracking, zero-copy transfer, admin APIs, and monitoring.
- Designed networked services around replication, message persistence, consumer coordination, and backend observability.
- Reported performance: 50,000+ messages/sec and support for 100+ concurrent consumers in project benchmarks.

VectorFlow – AI Vector Database Python 3.11, FastAPI, NumPy, Numba, HNSW
github.com/avishkar-004/vectorflow

- Built a vector database with HNSW approximate nearest-neighbor search, WAL persistence, consistent hash sharding, FastAPI APIs, and RAG pipeline integration.
- Used Numba optimization for search performance and designed persistence/sharding components for AI retrieval workloads.
- Reported support for 1M+ vectors, under-100ms search latency, and 95%+ recall in project benchmarks.

Experience

Software Developer Intern, OSMOS Jul 2025–22 May 2026

- Worked on backend services using Node.js and Python.
- Contributed to product development from scratch and built scalable application components.

Software Developer Intern, Core-Decimal Solutions Feb 2025–Aug 2025

- Built full-stack features using Vue.js, Express.js, Sequelize, and MySQL.
- Developed reusable frontend components and backend models for dynamic UI mapping.

Software Engineer Intern, Campus Credential Jun 2024–Aug 2024

- Built a web-based society management system with resident management, complaint tracking, event scheduling, and community features.

Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript, SQL
Frameworks/Tools: Spring Boot, FastAPI, React, Node.js, Express.js, Docker, Git, Linux, AWS, CMake, Maven
Databases: PostgreSQL, MySQL, MongoDB, Redis
Core Areas: Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design, Deep Learning, Computer Vision

Certifications and Achievements

- AWS Certified Cloud Practitioner, May 2026.
- CCNA: Introduction to Networks; Switching, Routing, and Wireless Essentials; Enterprise Networking, Security, and Automation.
- NPTEL: Programming in Modern C++, Introduction to Machine Learning.
- Saylor University: Introduction to Cryptography and Network Security.
- First place in Technodium 25; second place in Code Sprint Competition.